

DILLON COOLEY

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PROFESSIONAL SUMMARY

U.S. Navy instructor pilot, weapons and tactics instructor, and fleet experimentation lead with over 1,000 flight hours as mission commander, bringing expertise in maritime ISR, airborne mine countermeasures, and aircraft survivability systems and operations. Combines deep tactical aviation expertise with hands-on leadership in rapidly prototyping and integrating emerging technologies, including AI-enhanced sensors and resilient communications, into real-world operations. Complements this practical background with graduate research on distributed ISR and expeditionary operations in contested environments, paired with a track record of leading teams and developing technical solutions that improve effectiveness and accelerate capability adoption.

SKILLS / CERTIFICATIONS

Programming & Technical Stack: MATLAB, Simulink, Python, VS Code, Git

Defense & Tactical Software: Maven Smart System, FADE/MIST, Thresher, Bodhi, Joint Mission Planning System (JMPS), Joint Munitions Effectiveness Weapon Engineering System (JWS), Target Acquisition Weapons Software (TAWS), Chatsurfer, MINEnet Tactical

Military Credentials: Maritime ISR WTI, Mine Countermeasures WTI, High Altitude Aviation Training Site (HAATS), Survival, Evasion, Resistance, and Escape (SERE) School

FAA Licenses: Part 61 Commercial Pilot, Certified Flight Instructor (CFI/CFII), S-70 Type Rating, Part 107 Remote Pilot

Security Clearance: Top Secret / Sensitive Compartmented Information (TS/SCI)

EXPERIENCE

Helicopter Sea Combat Weapons School Pacific (HSCWSP) – LT (O-3), U.S. Navy

San Diego, CA

MH-60S Weapons & Tactics Instructor (WTI) / Fleet Experimentation Lead / AMCM & Aircraft Survivability SME

2024 – Present

- Coordinated and instructed advanced tactical training for senior flight crews while serving as standardization instructor and instrument check pilot, ensuring 100% compliance with operating procedures and safety protocols.
- Directed technical evaluations and rapid fielding of emerging defense technologies, including electronic warfare sensors and pLEO satellite communications integration; strengthened MH-60S command and control networks and validated new capabilities through joint exercises.
- Led modernization of airborne mine countermeasures training systems by integrating artificial intelligence into airborne mine countermeasures detection software, leveraging post-mission sensor data and simulation environments to improve detection performance and aircrew training fidelity.
- Served as primary technical authority for aircraft survivability systems, EO/IR/RF theory, and countertactics; analyzed classified threat data to update software-defined defensive systems and electronic warfare parameters across the MH-60 community.

Helicopter Sea Combat Squadron TWO ONE (HSC-21) – LTJG/LT (O-2/O-3), U.S. Navy

San Diego, CA

Mission Commander / Aircraft Commander / Assistant Operations Officer / Detachment Leader

2020 – 2024

- Executed over 1,000 flight hours as mission commander supporting combat logistics, maritime reconnaissance, and airborne mine countermeasures; qualified across all MH-60S and MQ-8B mission sets as functional check pilot, NATOPS instructor, and tactical mission lead.
- Optimized unit-level operations, scheduling, and resource allocation for a squadron of 500+ personnel and 15+ aircraft as assistant operations officer, increasing operational readiness and training efficiency.
- Led a 30-person cross-functional expeditionary detachment onboard LCS-18 for a six-month deployment to U.S. Third and Seventh Fleet areas of responsibility; coordinated secure tactical communications architecture for rotary-wing aviation assets during an eight-month LHD deployment across U.S. Third, Fifth, and Seventh Fleet areas of responsibility.

Naval Aviation Flight School – ENS (O-1), U.S. Navy

Pensacola, FL / Corpus Christi, TX / San Diego, CA

Student Naval Aviator / Fleet Replacement Pilot

2018 – 2020

- Completed an intensive technical flight training program, mastering aerodynamic principles, advanced navigation, and integrated avionics in high-performance aircraft. Achieved qualifications in multiple airframes, including the T-6B, TH-57, MH-60S, and MQ-8B, through rigorous academic study and in-flight training.

TECHNICAL PROJECTS

pLEO SATCOM & Resilient C2 Network Prototype (Professional)

- Identified critical beyond-line-of-sight C2 and full-motion video gaps for the MH-60S. Partnered with industry to prototype hardware and software solutions that resolved encryption, latency, and network integration challenges. Delivered training materials and demonstrated expanded resilient communications capability in major joint exercises (RIMPAC, Resolute Hunter, UNITAS).

Squadron Operations Optimization Tool (Personal / Side Project)

- Designed and prototyped a web application to automate and centralize daily/weekly/monthly operations tracking, training scheduling, and maintenance management for squadron departments. Intended to reduce administrative burden, improve data visibility/accuracy, and enable more efficient, data-driven planning and decision-making across operations, training, and maintenance teams.

EDUCATION

University of Staffordshire

Stoke-on-Trent, UK

Master of Science in Air and Space: Operations and Tactics

2024 – 2026

Thesis: *Countering A2/AD Through EABO: The Role and Feasibility of Distributed ISR in the Indo-Pacific*

United States Naval Academy

Annapolis, MD

Bachelor of Science in Aerospace Engineering

2014 – 2018